Predictions

1 Determine if the following predictions are extrapolations or interpolations:

a A prediction for the y-value when $x = 5$ is made from the data set below:

x	4	7	8	11	12	13	17	18	19	20
y	0	2	4	7	6	4	8	8	11	8

b A prediction for the y-value when $x = 33$ is made from the data set below:

x	37	54	58	59	43	55	60	38	64	35
y	72	53	26	21	73	47	12	102	10	112

c A prediction of $y = 95.69$ is made from the following data set using the line of best fit with equation $y = -0.07x + 96.18$:

x	19	10	1	7	14	11	2	5	17	8
y	94	94.4	97	96.4	94.4	97.8	94.9	95.9	96	94.4

d A prediction of $y = 72.77$ is made from the following data set using the line of best fit with equation $y = 1.26x - 57.01$:

x	93	57	86	97	78	96	68	69	54	92
y	51.2	25.4	38.9	58.6	38.2	60.8	26.3	28.5	5.4	92

2 Predict the value of y, using the given value of x and the line of best fit:

a $y = -8.71x + 6.79$; $x = 3.49$

b $y = 8.84x$; $x = 7.68$

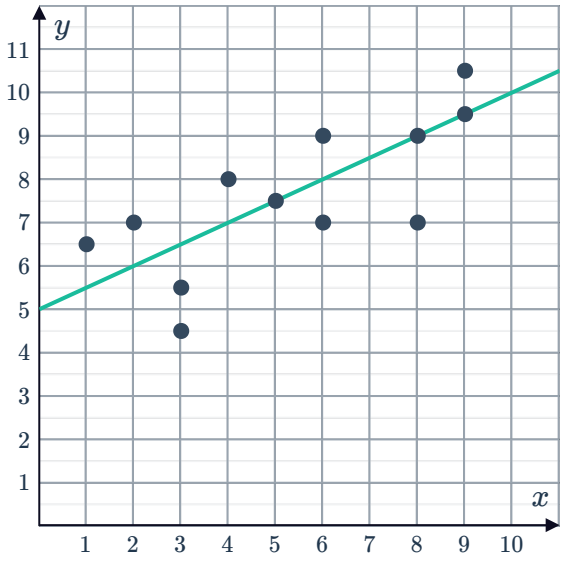
c $y = -0.84x - 0.19$; $x = -43.15$

d $y = 22.42x + 2.93$; $x = 0.26$

3 The equation $d = -0.58h + 51$ represents the line of best fit relating the air humidity, h, and the depth, d centimetres, of snow in an area.
Use the equation to determine the snow depth when the air humidity is 0.8.

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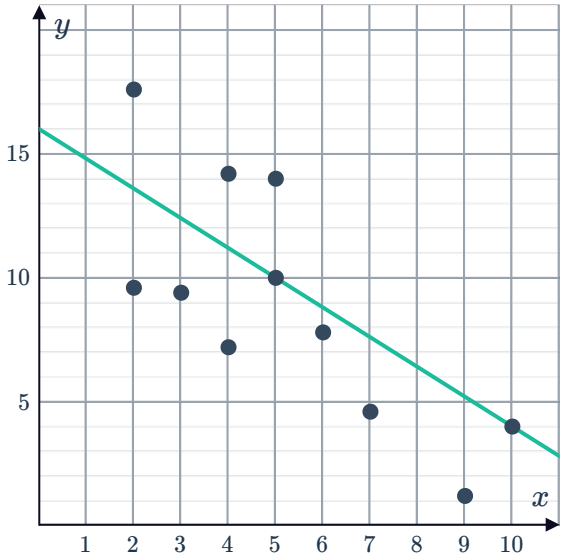
4 Consider the following scatter plot:



a Find the equation of the line of best fit.

b Use the line of best fit to approximate the value of y for $x = 6.9$.

5 Consider the scatter plot shown:



a Find the equation of the line of best fit.

b Predict the value of y when $x = 6.5$.

6 The data below has a correlation coefficient $r = 0.88$ and a line of best fit of $y = 0.85x + 1.54$.

x	20	18	12	11	10	3	16	8	6	5
y	17	20	9	14	8	5	14	12	3	6

a Predict the value of y when $x = 14$.

b Is this prediction reliable? Explain your answer.

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7 The data below has a correlation coefficient $r = 0.45$ and a line of best fit of $y = 0.22x + 0.7$.

x	23	81	11	44	50	91	51	95	53	82
y	1	35.2	1	12.8	23	1.7	21.2	35	1	2.9

a Predict the value of y when $x = 143$.

b Is this prediction reliable? Explain your answer.

8 The data below has a correlation coefficient $r = 0.57$ and a line of best fit of $y = 0.27x - 0.62$.

x	23	81	11	44	50	91	51	95	53	82
y	1	35.2	0.8	12.8	23	1.7	21.2	35	1	19

a Predict the value of y when $x = 145$.

b Is this prediction reliable? Explain your answer.

9 The data below has a correlation coefficient $r = -0.55$ and a line of best fit of $y = -0.54x + 123.74$.

x	19	67	2	66	97	88	59	75	51	14
y	83.2	46.6	143.6	138.8	68.1	92.4	81.2	70	103.8	119.2

a Predict the value of y when $x = 55$.

b Is this prediction reliable? Explain your answer.

10 The data below has a correlation coefficient $r = -0.99$ and a line of best fit of $y = -3.04x + 91.11$.

x	70	79	100	67	63	61	58	53	95	81
y	-115	-154	-219	-104	-93	-103	-83	-79	-194	-155

a Predict the value of y when $x = 45$.

b Is this prediction reliable? Explain your answer.

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Applications

11 One litre of gas is raised to various temperatures and its pressure is measured as shown:

Temperature (K)	300	302	304	308	310	312	314	316	318
Pressure (Pa)	2400	2416	2434	2462	2478	2496	2512	2526	2546

The data has been graphed along with a line of best fit:

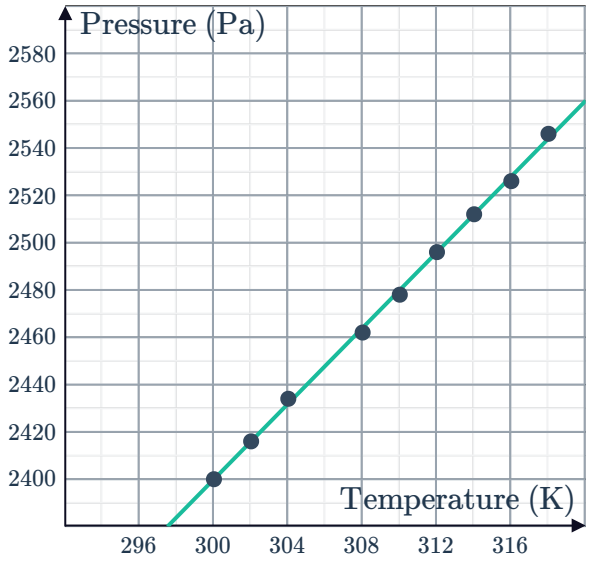
a The pressure was not recorded when the temperature was 306 K.
Is it reasonable to use the line of best fit to predict the pressure? Explain your answer.

b Predict the pressure when the temperature is 306 K.

c State whether it is reasonable to use the line of best fit to predict pressure for the following ranges of temperatures:

i $[300, 320]$ ii $[300, 600]$

iii $[0, 320]$ iv $[280, 340]$



12 An ice cream shop records the number of ice creams sold and the maximum temperature of each day for two weeks:

Temp (°C)	35	29	34	33	25	25	34	34	32	35	33	28	33	32
Sales	70	45	71	67	30	28	65	75	60	74	63	35	62	65

The data has been graphed along with a line of best fit:

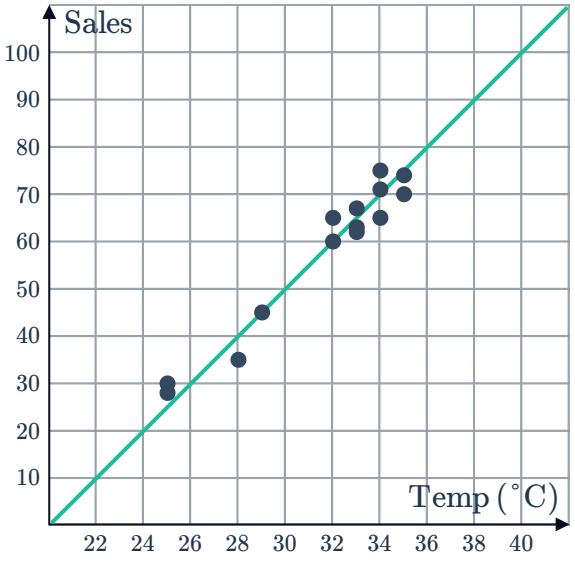
a Predict the number of ice creams sold if the maximum temperature for a day is 38°C.

b Predict the maximum temperature if 10 ice creams were sold.

c State whether it is reasonable to use the line of best fit to predict ice cream sales within each of the following range of temperatures:

i $[0, 40]$ ii $[20, 40]$

iii $[20, 60]$ iv $[0, 60]$



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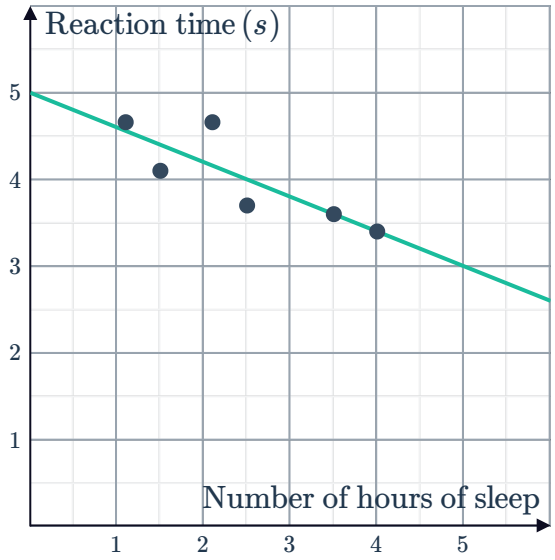
13 Scientists conducted a study to see people's reaction times after they've had different amounts of sleep. The results are recorded in the table:

Number of hours of sleep (x)	1.1	1.5	2.1	2.5	3.5	4
Reaction time in seconds (y)	4.66	4.1	4.66	3.7	3.6	3.4

The data has been graphed along with a line of best fit:

a Predict the reaction time for someone who has slept for 5 hours.

b Predict the number of hours someone sleeps if they have a reaction time of 4 seconds.



14 The number of fish in a river is measured over a five year period:

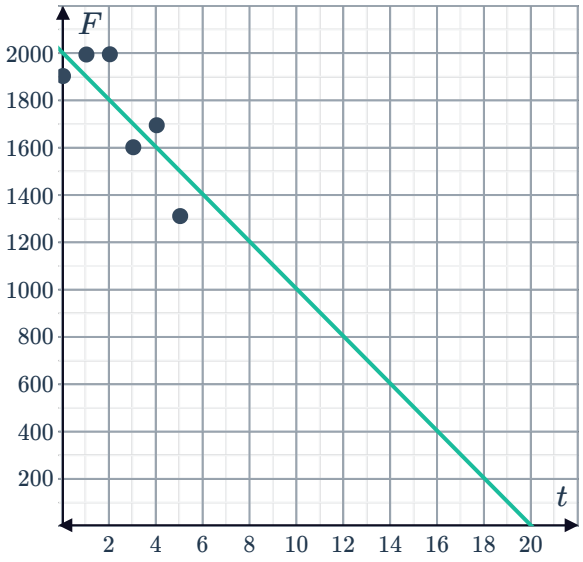
Time in years (t)	0	1	2	3	4	5
Number of fish (F)	1903	1994	1995	1602	1695	1311

The data has been graphed along with a line of best fit:

a Predict the number of years until there are no fish left in the river.

b Predict the number of fish remaining in the river after 7 years.

c According to the line of best fit, how many years are there until there are 900 fish left in the river?



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15 A plane's altitude (A) is measured at several times (t) during its descent:

Time (t seconds)	0	200	400	1700
Altitude (A metres)	9000	7815	7092	593

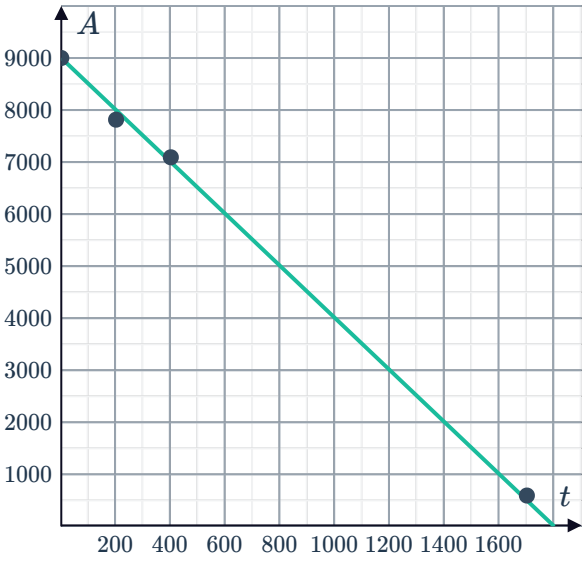
The data has been graphed along with a line of best fit:

a Predict the altitude of the plane 100 seconds into the descent.

b Predict the altitude of the plane 500 seconds into the descent.

c For how many seconds has the plane been descending when it is at an altitude of 7500 metres?

d How many seconds did the plane take to descend to the ground?



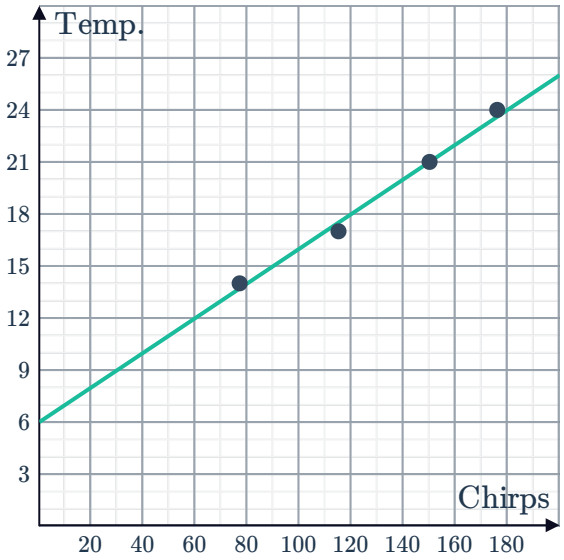
16 Chirping crickets can be an excellent indication on how hot or cool it is outside. Different species of crickets have different chirping rates but for a particular species the following data was recorded:

Number of chirps per minute	77	115	150	176
Temperature (°C)	14	17	21	24

a What is the temperature when the crickets make 140 chirps each minute?

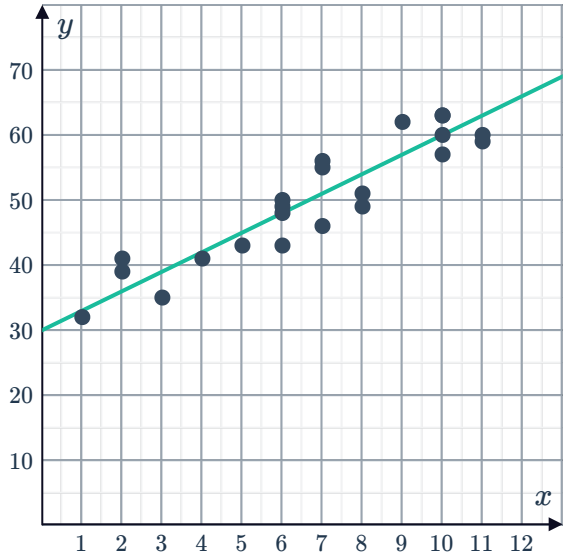
b How many chirps per minute will the crickets make if the temperature is 27?

c How many chirps are the crickets making each minute if the temperature is 19°C?



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17 Several cars underwent a brake test and their age, x, in years was measured against their stopping distance, y, in metres. The scatter plot shows the results and a line of best fit that approximates the positive correlation:



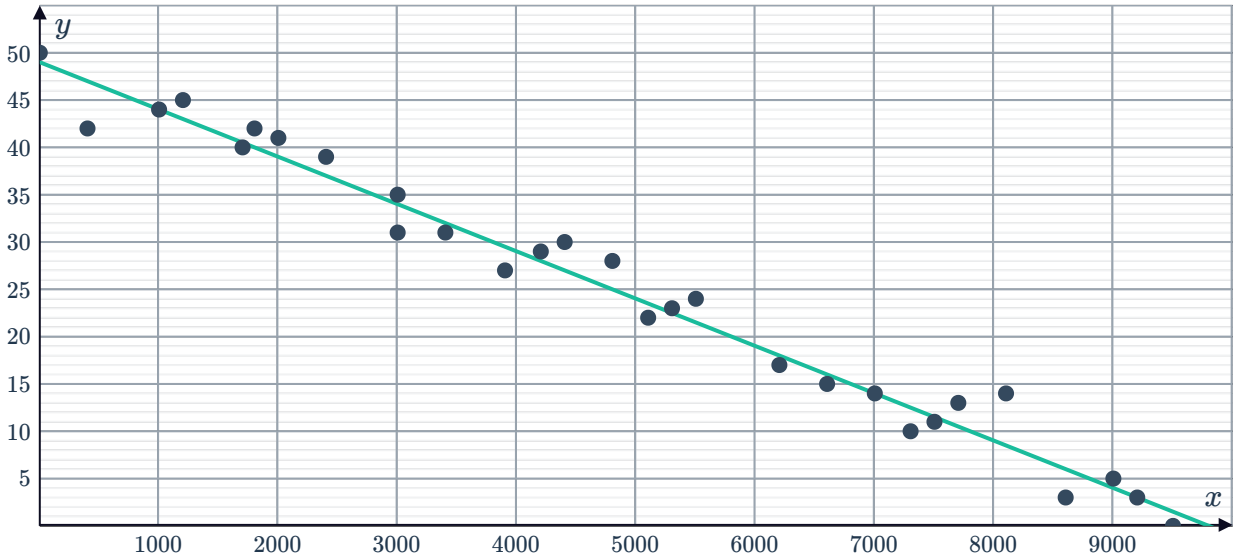
a Predict the stopping distance of a car that is 6 years old.

b Find the equation of the line of best fit.

c Use the line of best fit to estimate the stopping distance of a car that is 7.5 years old.

d Is the predicted value in part (c) reliable? Explain your answer.

18 The distance in kilometers, x, of several locations from the equator and their temperature, y in $^\circ\mathrm{C}$, is measured and graphed on the following scatter plot:



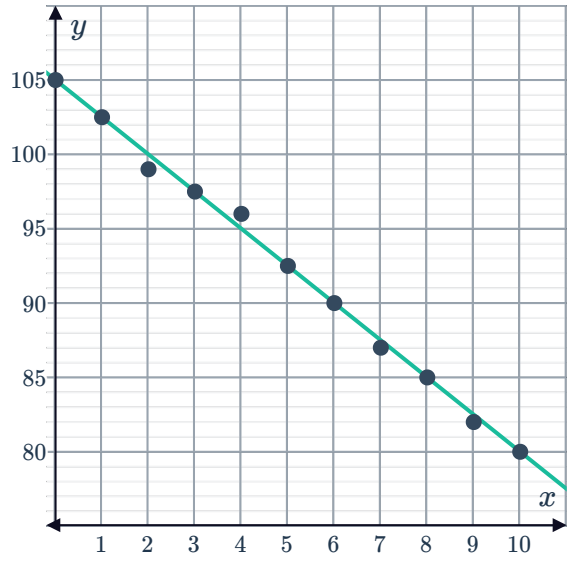
a Find the equation of the line of best fit shown.

b If your distance from the equator is $4000\ \mathrm{km}$, estimate the temperature, y, using the equation in the previous part.

c Is the predicted value in part (b) reliable? Explain your answer.

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19 The depth a diver, x, has descended below the surface of the water is plotted against her lung capacity, y:



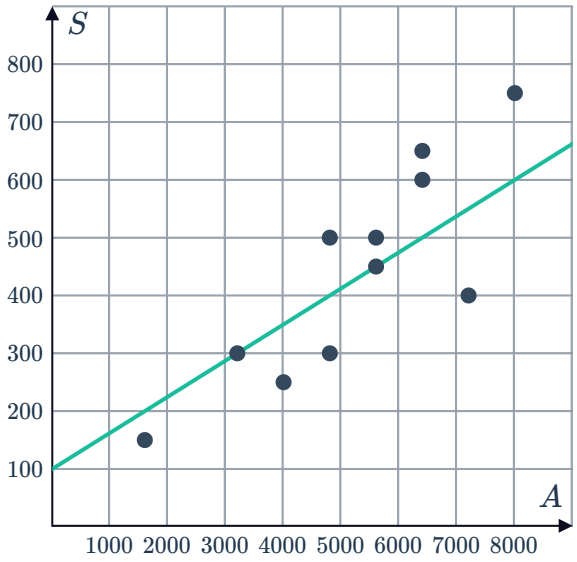
a Does the line of best fit have a positive or negative gradient?

b Find the gradient of the line.

c Find the equation of the line of best fit.

d Use the line of best fit to estimate the lung capacity, y, at a depth of 4 metres.

20 A car company looked at the relationship between how much it had spent on advertising and the amount of sales each month over several months. The data has been plotted on the scatter graph and a line of best fit drawn:

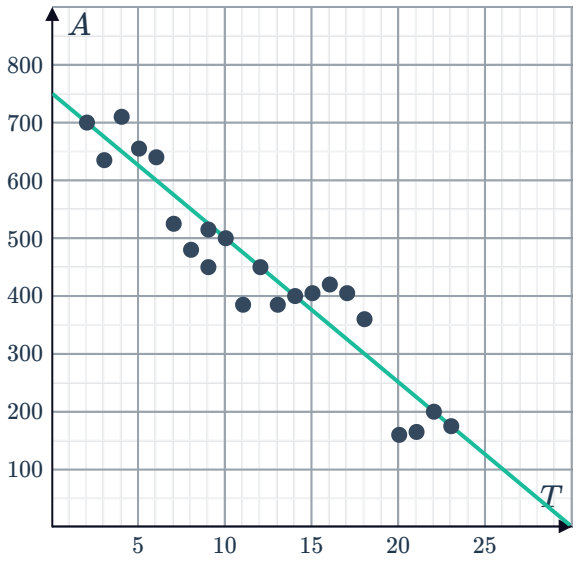


a Two points on the line are $(3200, 300)$ and $(5600, 450)$. Find the gradient of the line of best fit.

b The line of best fit can be written in the form $S = mA + c$, where m is the gradient, c is the vertical intercept, S is the money spent on sales in thousands of dollars, and A is the advertising costs. Find the value of c, the vertical intercept of the line.

c Use the line of best fit to estimate the number of sales next month if $\$4800$ is to be spent on advertising.

21 A line of best fit has been drawn to approximate the relationship between sea temperature, T in $^\circ\text{C}$, and the area of healthy coral, A in hectares, in a particular location. Two particular points, $(2, 700)$ and $(23, 175)$, lie on the line.



a Find the equation of the line of best fit.

b Using the line of best fit, find the area A of coral expected when the sea temperature is 26°C.

c Is the predicted value in part (b) reliable? Explain your answer.

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Ducks	1	2	3	4	5	6	7	8
Eggs	4	11	14	21	24	31	34	41

a Construct a scatter plot for this data on a number plane.

b Sketch the line of best fit for the data, given that the line passes through $(5, 25)$ and $(-1, -5)$.

c Find the gradient of the line of best fit.

d Find the y-intercept of the line of best fit.

e Find the equation of the line of best fit.

f Use the equation to find the number of eggs laid by 30 ducks.

23 The table shows data on the number of kilograms of litter collected each week in a national park x weeks after the park managers started an anti-littering campaign:

Weeks (x)	1	2	3	4	5	6	7
Kilograms of litter collected (y)	3.9	3.7	3.5	3.1	3.1	2.9	2.7

a Construct a scatter plot for this data.

b Sketch a line of best fit on your scatter plot.

c Find the equation for your line of best fit.

d Use your equation of the line of best fit to find the number of kilograms of litter collected 8 weeks after the start of the anti-littering campaign.

e Is the predicted value in part (d) reliable? Explain your answer.

24 The table shows the number of people who went to watch a movie x weeks after it was released:

Weeks (x)	1	2	3	4	5	6	7
Number of people (y)	17	17	13	13	9	9	5

a Construct a scatter plot for this data on a number plane.

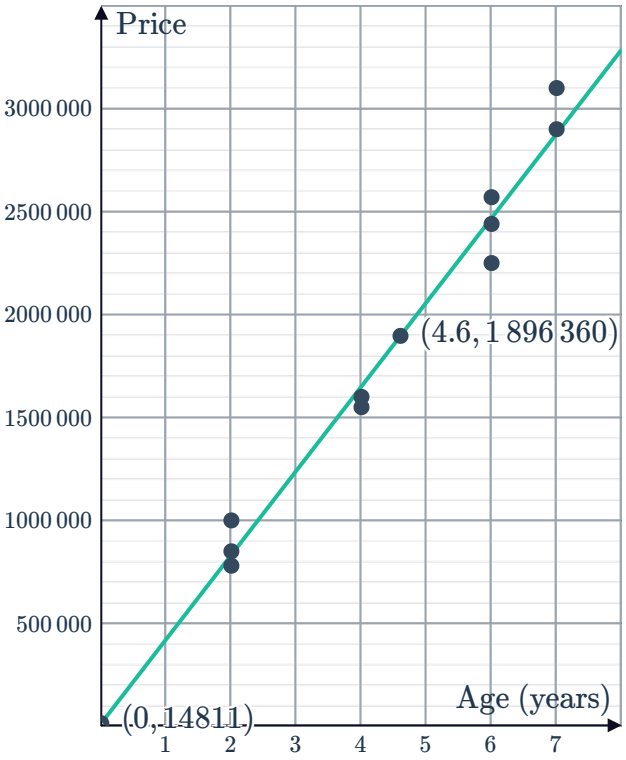
b Sketch a line of best fit on your scatter plot.

c Find the equation of your line of best fit.

d Use your equation to find the number of people who went to watch the movie 10 weeks after it was released.

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25 The following scattergraph shows the value of various 4 bedroom, 2 bathroom homes in a new suburb:



a State the y-intercept.

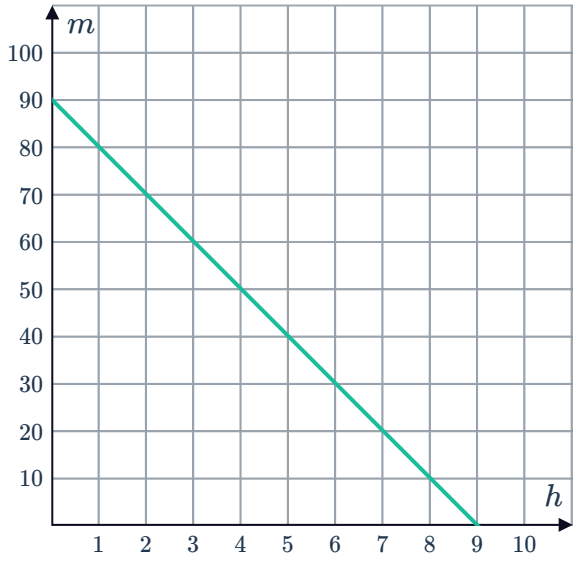
b Explain what the y-intercept represents.

c Does the interpretation in the previous part make sense in this context? Explain your answer.

d Use the graph to estimate the age of a house with an average value of $\$2\,000\,000$. Give your answer in years to one decimal place.

e Use the graph to estimate the average price of a house with age 3 years.

26 The number of hours spent watching TV each evening, h, is measured against the percentage results, m, achieved in the Economics exam. The line of best fit for the resulting data is shown:



a Explain what the y-intercept represents.

b Does the interpretation in the previous part make sense in this context? Explain your answer.

c Would it be reasonable to use the line of best fit to predict average mark of students who watch more than 10 hours of TV?

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27 The price of various second-hand Mitsubishi Lancers are shown in the table below:

Age	1	2	0	5	7	4	3	4	8	2
Value (in thousands of dollars)	16	13	21.99	10	8.6	12.5	11	11	4.5	14.5

The line of best fit for the data is shown below:



a State the y-intercept to the nearest thousand.

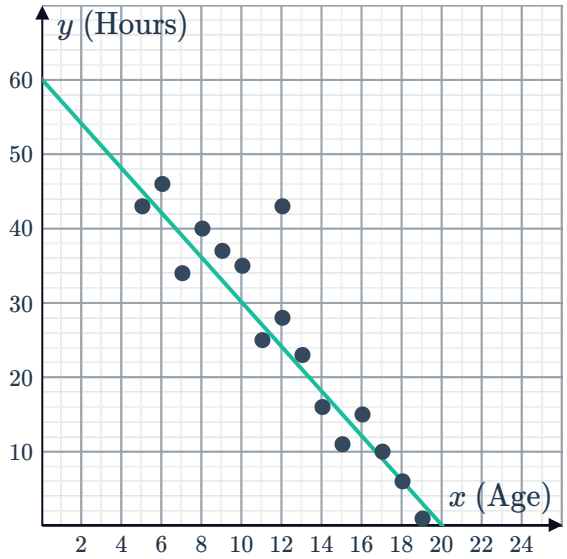
b Explain what the y-intercept represents.

c Does the interpretation in the previous part make sense in this context? Explain your answer.

d At what age, to the nearest year, would a car be worth \$0?

e Can we use this line of best fit to find the value of a car that is older than 11 years? Explain your answer.

28 A number of people were asked how many hours each week they spend on the internet. Their results were graphed against their age in the scatter plot and a strong negative correlation was observed. A line of best fit has been drawn along with the data:



a Find the equation for the line of best fit in the form $y = mx + b$.

b Consider the response which is an outlier. According to the line of best fit, what should their usage be?

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29 The following table shows the temperature of a cooling metal versus time:

Minutes	1	2	3	4	5	6
Temperature	27	27	23	23	19	19

a Plot the points from the table on a number plane.

b Sketch a line of best fit on your scatter plot, given that the line passes through $(5, 20)$ and $(3, 24)$.

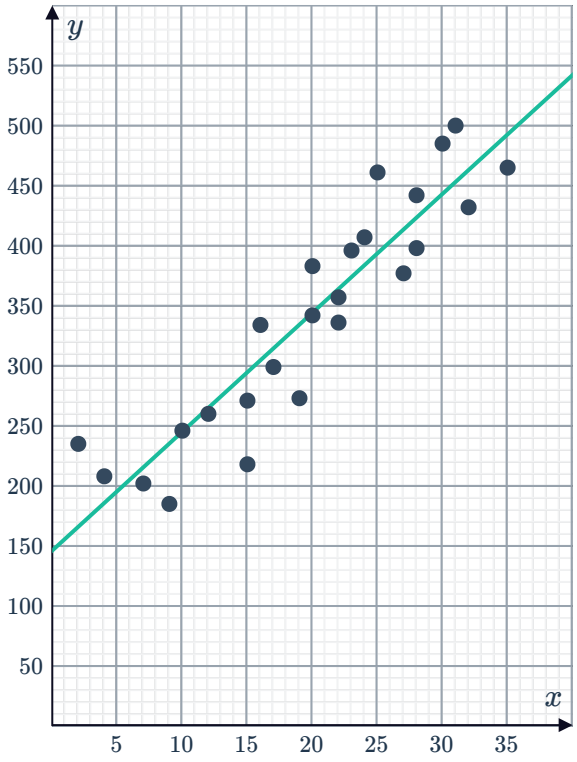
c Find the gradient of the line of best fit.

d Find the y-intercept of the line of best fit.

e Find the equation of the line of best fit.

f Use the equation to find the number of minutes required to reach the temperature of 15°C in terms of x.

30 The fleet manager for the Australian Automotive Association wants to estimate how car maintenance costs, C (in hundreds of dollars), are related to the distance, K (in thousands of kilometres), driven each year.
The data collected and the line of best fit are shown on the scatter plot below:



a The equation of the least-squares line for the graph is $C = 9.9K + 146.2$. Interpret what the coefficient of K represents in this context.

b Explain what the constant term in the equation represents in this context.

c Is this interpretation of the constant term reasonable given the context? Explain your answer.

d Use the least-squares equation $C = 9.9K + 146.2$ to predict the annual maintenance cost for a car that is driven $36\,000$ km per year.

e Is this predicted value reliable? Explain your answer.

f Is there any reason to believe that there is a causal relationship between distance driven and maintenance costs? Explain your answer.

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